

The Challenge

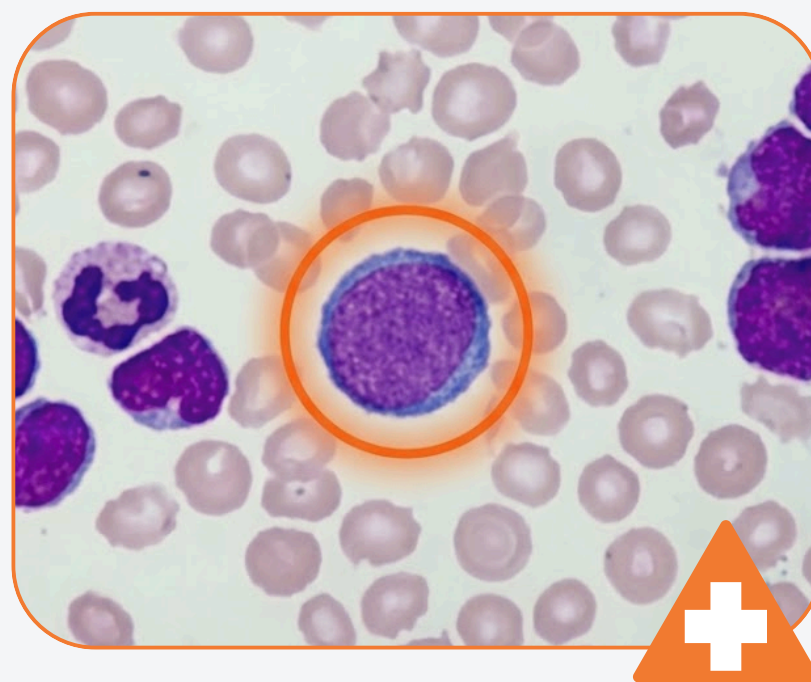
[Traditional Diagnosis] → [AI-Based Detection]

- | | |
|---------------|------------|
| ✗ Slow | ✓ Fast |
| ✗ Subjective | ✓ Accurate |
| ✗ Error-prone | ✓ Reliable |

Every missed leukemia case could **cost a life !!**

+92%

Leukemia Recall



AI-Based Blood Smear Analysis

+10,000

Blood Smear Images

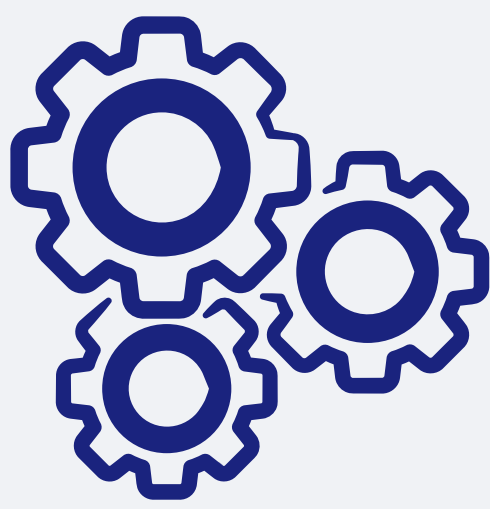
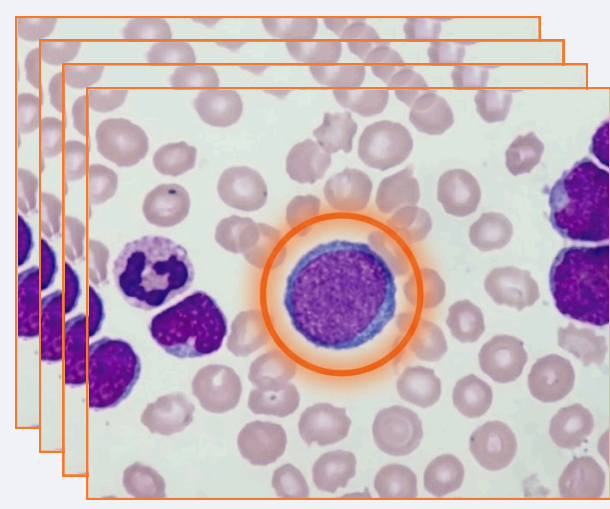
Abstract



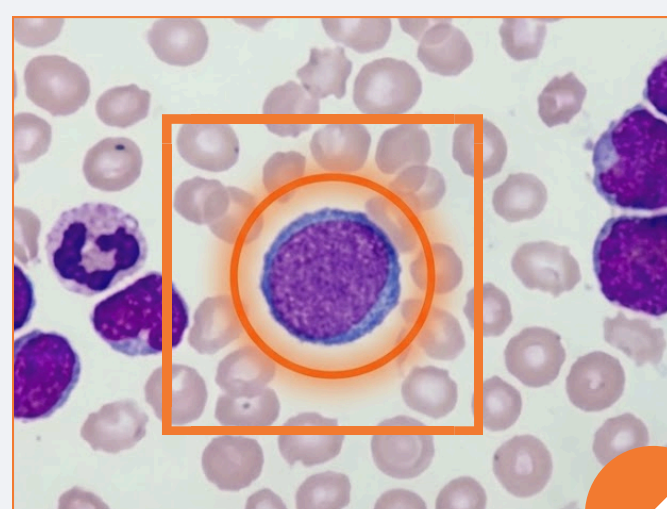
Leukemia diagnosis through manual blood smear examination is time-consuming and prone to human error. This study proposes a hybrid deep learning framework for blood cell classification and leukemia detection using medical images. The proposed approach achieved high classification performance and reliable leukemia detection results.

Methodology

Blood Smear Images Data Pre-processing Classification & Output

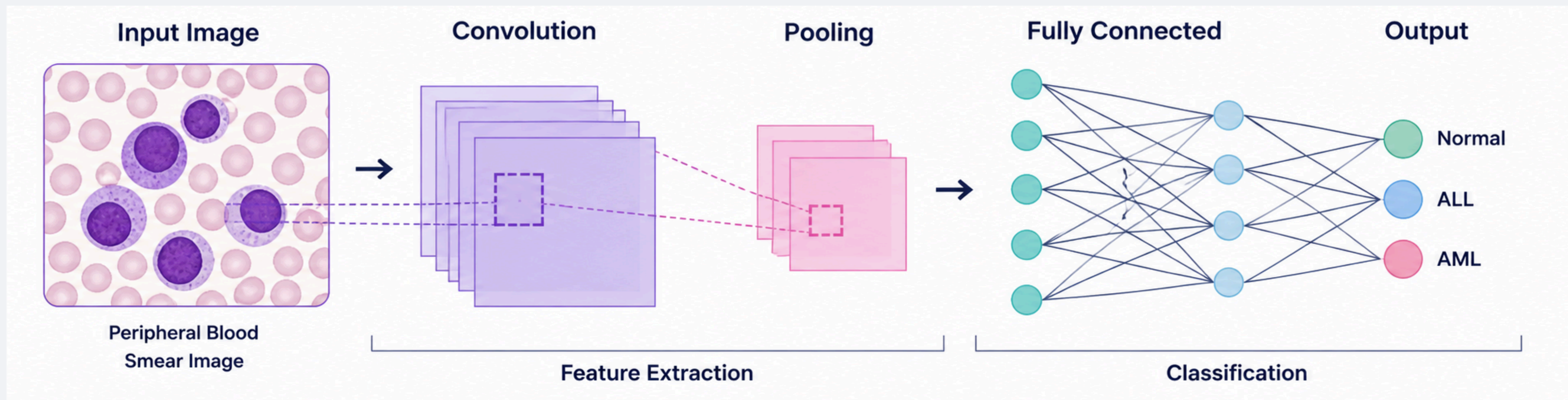


- Contrast Enhancement (HSI)
- Normalization
- Resizing
- Data Augmentation



(Normal | ALL | AML)

Deep Learning Models



(EfficientNetV2 + ConvNeXt)

Results

+92%

Recall Score



The hybrid model achieved high recall and balanced performance.

Future Work

What's next ?

Advancing leukemia detection beyond current limits

- Larger datasets
- Better accuracy
- Real-time detection
- Clinical use

Blood Cell Classification

+96%

Accuracy

+96%

F1

Conclusion

The proposed hybrid model achieves high accuracy and reliable leukemia classification.

